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SIMATS SCHOOL OF ENGINEERING

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B.E./ B.Tech DEGREE EXAMINATION – AUG 2026 MODEL PRACTICAL EXAMINATIONS CSA51 - CRYPTOGRAPHY AND NETWORK SECURITY

SET 1

1. Use the poly alphabetic cipher to encrypt the message "SHE IS LISTENING TO THE CLASS" with key "CRYPTO".
 - a. Encrypt the above plaintext
 - b. Apply Decryption technique to obtain the given text.
2. Write a C program for RSA public-key encryption scheme, each user has a public key, e , and a private key, d . Suppose Bob leaks his private key. Rather than generating a new modulus, he decides to generate a new public and a new private key. Is this safe? Justify with implementation.
3. Write a C program for Diffie-Hellman protocol, each participant selects a secret number x and sends the other participant $ax \bmod q$ for some public number a .

A) What would happen if the participants sent each other xa for some public number a instead? Give at least one method Alice and Bob could use to agree on a key.

B) User A (Alice) & B (Bob) exchange the key using Diffie Hellman alg. Assume $\alpha=8$
 $q=33$ $XA=3$ $XB=2$. Find YA , YB , K .
4. Encrypt "cat" using Hill Cipher with key
 - (3 2 1)
 - (4 3 2)
 - (1 2 4)

And apply the decryption process at the receiver's end to retrieve the original message

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Practical Model Examination.

1.) Polyalphabetic Cipher

Aim:

to implement the polyalphabetic cipher to encrypt the message with key.

Algorithm:

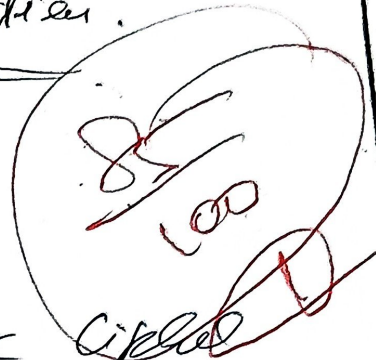
(a) i) Firstly we have to read the plaintext (SHE IS LISTENING TO THE CLASS) and key (CRYPTO).

ii) Repeat the key until it matches the length of the plaintext.

iii) Convert the letters into numbers (A=0, B=1, ..., Z=25).

iv) $C = (P + K) \bmod 26$.

v.) The result is cipher text.



b.) $\Rightarrow$ i) Read the ^{ciphertext} ~~plaintext~~ and key.

ii) Repeat the key value to match the ciphertext length.

iii) Convert letter to number ($A=0, B=1, \dots, Z=25$).

$$iv) P = (C - K + 26) \bmod 26.$$

v.) The result is plaintext.

Pseudo Code:

i) MSG = "SAP IS LISTENING TO THE CLASS"

ii) KEY = "CRYPTO"

$$iii) C = (P + K) \bmod 26$$

iv) $P =$ Plaintext

$K =$ Key

v.) $A=0, B=1, \dots, Z=25.$

RSA Public-key Encryption:

Sim:

To implement RSA public key encryption using a C program.

Algorithm:

i) Firstly we have to choose two prime numbers (p & q).

$$ii) n = p \times q.$$

$$iii) \phi(n) = (p-1)(q-1)$$

$$iv) 1 < e < \phi(n) \text{ and } \gcd(e, \phi(n)) = 1.$$

$$v) d = e^{-1} \text{ mod } \phi(n)$$

$$vi) \text{Public Key} = (e, n)$$

$$vii) \text{Private Key} = (d, n)$$

Encryption:

i) Read the plaintext

$$ii) C = M^e \text{ mod } n$$

iii) Result is ciphertext.

Decryption:

- i) Read ciphertext
- ii) $m = c^d \text{ mod } n$
- iii) Return it plaintext.

3) Diffie-Hellman:

Aim:

to implement the Diffie-Hellman using C-program.

Algorithm:

i) Firstly choose any prime q and primitive root a .

ii) Alice select private key x_A .

iii) Bob select private key x_B .

iv) Alice $\Rightarrow Y_A = a^{x_A} \text{ mod } q$

v) Bob $\Rightarrow Y_B = a^{x_B} \text{ mod } q$

vi) Alice sends Y_A to Bob.

vii) Bob sends V_B to Alice.

viii) $K = V_B^A \times A \Rightarrow \text{Alice}$

ix) $K = V_A^B \times B \Rightarrow \text{Bob}$

x.) Both get same secret key (or) private key K .

4.) Hill Cipher:

Aim:

to implement Hill Cipher using C program.

Algorithm:

Encryption:

- i) Read the plaintext and key properly.
- ii) ~~convert~~ letters to numbers ($A=0$ to $Z=25$)
- iii) Divide plaintext into blocks matching the key.
- iv) multiply the key by each plaintext
- v) multiply each element with mod.
- vi) convert back it
- vii) Result is ciphertext.

Decryption:

- i) Find the inverse of key mod 26.
 - ii) Convert ciphertext into numbers.
 - iii) Multiply the number key matrix with each ciphertext.
 - iv) Result mod 26.
 - v) Convert back to.
 - vi) Result is plaintext.
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